

Unsupervised Health Indicator Fusing Time and Frequency Domain Information and Its Application to Remaining Useful Life Prediction

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Abstract—The prediction of component remaining useful life (RUL) is essential in making an appropriate maintenance plan for equipment. Constructing a reliable health indicator (HI) is crucial for RUL prediction. HI can be generated by quantifying distribution discrepancies. Most existing methods construct HIs based on the time domain, whereas in certain cases, time-domain data contain fewer degradation characteristics than frequency-domain data. To enhance the applicability and quality of HIs under different conditions, this article presents a novel unsupervised approach for generating HI from both the time and frequency domains. Considering the frequency-domain data characteristics of mechanical vibration signals, an exponential mixture model (EMM) is first applied to extract the frequency-domain distribution characteristics. Furthermore, a Gaussian mixture model (GMM) is used to mine time-domain distribution characteristics. Subsequently, a distribution contact ratio metric (DCRM) is employed to respectively generate the time and frequency domain HIs by quantifying the discrepancies between baseline distribution and data distributions at different degradation moments. The final HI is constructed by weighting the time and frequency domain HIs. RUL prediction is achieved using the Proposed-HI and a variant of recurrent neural network. Finally, the efficiency and superiority of this approach are validated using multiple gear life-cycle datasets, and the presented HI exhibits a higher RUL prediction accuracy than classical and advanced unsupervised HIs.

Index Terms—Distribution estimation, health indicator (HI), mixture model, remaining useful life (RUL) prediction, unsupervised learning.

I. INTRODUCTION

MECHANICAL equipment, such as automobile engines, pumps, aircraft, and wind turbines, inevitably degrades under complex working conditions and severe external environments [1], [2]. To guarantee the normal and efficient operation of such equipment and avoid the occurrence of serious catastrophic accidents, it is essential to robustly and reliably predict the remaining useful life (RUL). Using the estimated RULs, appropriate control strategies and maintenance decisions can be made to prevent equipment downtime [3], [4], [5]. As critical power transmission components of such

equipment, gears are prone to failure [6], [7]. It is of great engineering value to quantify the degradation process of gears and estimate their RULs. Therefore, this work explores a novel health indicator (HI) construction method with superior performance and its application to the RUL prediction of gears.

RUL prediction approaches, which are one of the fundamental content of prognostics and health management (PHM), can be categorized as physics-based approaches and data-driven approaches [8], [9]. The former can determine accurate RULs based on certain specific physical models. However, such methods require a comprehensive understanding of degradation mechanisms and evolution laws. This limits their application to complex mechanical equipment due to the complex failure mechanisms and working environments met in practical engineering applications. Data-driven methods, which can directly extract degradation information from monitoring data by machine learning technologies, have been explored to implement the RUL estimation of sophisticated mechanical equipment.

Nowadays, different data-driven methods have been explored to predict the RUL of machines via certain neural networks. For example, Xiang et al. [10] proposed a novel deep learning (DL) network for RUL prediction of machines, which is able to adaptively mine abstract features from multidimensional monitoring data. Based on the recurrent neural networks, an ensemble RUL prediction network was developed by Hu et al. [11] to estimate the accurate RUL of aircraft engines. Chen et al. [12] introduced a long short-term memory (LSTM) network-based method with attention mechanism for the RUL prediction of machines. Ma and Mao [13] designed a convolution-based LSTM model to predict machine's RUL via information mined from monitoring data. Chen et al. [14] developed a variant of recurrent neural networks to obtain the life of servo motor, considering the sequential information in the attention gate. It is noteworthy that the current data-driven RUL prediction methods mainly comprise three steps: monitoring data collection, feature extraction (HI construction), and RUL prediction (time series or regression prediction).

As a significant and indispensable step in the data-driven RUL prediction approach, the feature extraction (HI construction) step is used to generate an HI that can dynamically and quantitatively evaluate the machine deterioration degree [15]. In addition, an appropriate HI is conducive to acquiring more accurate prediction results. Thus, it is essential to develop an

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approach for automatically constructing HI from monitoring data. The constructed HI can quantitatively characterize the deterioration process of machines and be applied to RUL prediction combined with the failure threshold. Numerous works on the construction of machine HIs have been developed. These HIs can be classified as physical HIs (PHIs) and virtual HIs (VHIs) [16]. As one of the commonly utilized PHIs, the root mean square (rms) is capable of representing the degradation trend of machines. For instance, Li et al. [17] utilized rms as an HI to estimate the bearing's RUL. Huang et al. [18] applied rms to acquire the RUL of bearings using an adaptive stochastic process model. Although rms has an explicit physical meaning, it cannot adequately capture the actual degradation of machines since it relies solely on a single time-domain feature.

Several popular dimension reduction methods have been applied to construct VHIs able to fully represent the actual degradation process of machines. For instance, Zhao et al. [19] adopted principal component analysis (PCA) to generate HIs for bearings based on time and frequency domain data. Chen et al. [20] introduced a variant of PCA methods to reflect the nonlinear degradation process of mechanical equipment. Based on the extracted statistical features in time and frequency domains, Qin et al. [21] utilized isometric mapping (ISOMAP) to acquire an improved gear HI by fusing them. Nevertheless, the performance of the above VHIs may be affected by certain inapplicable features. In addition, such dimension reduction methods cannot be directly applied to the construction of HIs from raw monitoring data.

DL approaches have been implemented to construct VHIs from raw monitoring data since they can overcome the limitations imposed by dimension reduction methods. DL-based VHIs can be established via two distinct ways: supervised and unsupervised learning. Several HI construction methods using supervised learning have recently been explored for generating bearing's HIs. For instance, Guo et al. [22] used a DL-based framework to adaptively construct HIs of bearings by a linear HI label. Cheng et al. [23] employed a convolutional neural network to generate HIs of rolling bearings through a nonlinear HI label. Notably, these methods used designed HI labels to construct HIs. However, due to the complex and volatile working conditions, the actual degradation trend may deviate from these HI labels. Consequently, the adoption of such HI labels may result in an inaccurate degradation trend, rendering the constructed HIs incapable of accurately representing the actual health status of machines. On the contrary, unsupervised HI construction methods are capable of constructing HIs without any HI labels and expert knowledge. Therefore, such methods have attracted considerable attention in the HI construction field and have been widely investigated in recent years. For example, Li et al. [24] proposed an unsupervised constraint model to construct HI with a designed loss function. In another study, Qin et al. [25] built a novel variant of autoencoders for generating HIs of machinery with trend-constrained function. Guo et al. [26] introduced an approach to construct HIs for health monitoring of machines. Nevertheless, current DL-based unsupervised HI construction methods require sufficient monitoring data to be trained, and in practical

engineering applications, it is challenging to gather enough monitoring data to train the HI construction approaches.

The Gaussian mixture model (GMM) is a commonly used unsupervised clustering method that has been applied to construct HIs from the data distribution perspective. It can construct machine HIs using limited life-cycle data. Several GMM-based HI construction methods have been proposed. For instance, Yu [27] developed an adaptive GMM for machine tool condition monitoring. He and Guo [28] introduced an improved GMM for constructing bearing HIs in an unsupervised learning manner. Zhou et al. [29] presented an unsupervised HI construction model for building bearing's HIs using raw monitoring data and combining GMM and the designed distribution contact ratio metric (DCRM). The existing research on GMM-based HIs indicates that GMM can estimate the unknown distributions of run-to-failure data by combining finite Gaussian distributions. Subsequently, the HI is generated by measuring the discrepancy between the distribution of health data and that of degradation data at a time. GMM is capable of estimating data distributions from monitoring data in the time domain. However, time-domain signals cannot fully capture the degradation process of mechanical equipment. In certain cases, frequency-domain signals are able to better characterize the degradation process. Therefore, it is essential to extract degradation information from the frequency domain. It is well known that frequency-domain data can be obtained via fast Fourier transform (FFT), and the obtained FFT data contain the degradation information of machines. As time elapses, the distribution of FFT data changes, and the data distribution in the health stage becomes different from that in the degradation stage. Thus, the discrepancy between the baseline distribution of health FFT data and the distribution of life-cycle FFT data can be utilized as HI to describe the deterioration trend of machinery.

According to a series of distribution tests, the frequency-domain data of mechanical vibration signals mainly contain exponential distribution features. Therefore, an exponential mixture model (EMM) is introduced to extract the distribution characteristics from the frequency domain. To enhance the applicability and performance of HI in different cases, GMM is also applied to mine the distribution characteristics from the time domain. Subsequently, DCRM is employed to generate HIs in the above domains by measuring the discrepancies between the baseline distribution (healthy data) and the data distributions at different moments of degradation. Based on the HI monotonicity, the final HI is constructed by linearly weighting the obtained HIs. The Proposed-HI based on GMM, EMM, and DCRM can comprehensively describe the deterioration trend of machinery from various perspectives. Accurate RUL prediction results can be obtained via the constructed HI and time series-based life estimation approach. The experimental results on gear RUL prediction demonstrate the effectiveness of the developed HI construction approach. The primary contributions of this study are summarized as follows.

- 1) A novel HI construction method that takes both the time and frequency domains into consideration is proposed. By analyzing the characteristics of frequency spectra of machine vibration signals, EMM is first applied

to extract their distribution characteristics. Meantime, GMM is employed to mine the distribution characteristics of time-domain signals.

- 2) With the distribution characteristics of frequency and time domains, DCRM is used to obtain the frequency- and time-domain HIs. Considering the monotonicity of two HIs, a new method for linearly weighting the frequency-domain and time-domain HIs is proposed to generate the final HI.
- 3) A time series-based RUL prediction approach is applied to estimate RUL using Proposed-HI. The effectiveness and superiority of the Proposed-HI construction method are verified by the experimental results for several gear datasets. Finally, Proposed-HI is more appropriate to RUL prediction compared to the classical and advanced unsupervised HIs.

The remaining structure of this work is as follows. Section II presents the presented HI construction approach using GMM and EMM. Section III illustrates the effectiveness and superiority of the presented method using several gear datasets. Section IV demonstrates the application of constructed HI in gear RUL prediction. Section V summarizes this work.

II. PROPOSED-HI CONSTRUCTION APPROACH

The GMM is first introduced to extract time-domain distribution features from the Gaussian distribution perspective. Subsequently, the EMM is developed to mine frequency-domain distribution features from the exponential distribution perspective. The Proposed-HI construction approach is developed by using GMM, EMM, and DCRM. Finally, the HI is generated by the proposed method.

A. Gaussian Mixture Model

As a popular clustering method, GMM has been commonly applied in various fields, e.g., data classification [30], anomaly detection [31], image segmentation [32], and image recognition [33]. The overall density function of GMM is formed via a weighted combination of finite Gaussian density functions. The overall density function can be defined as follows:

$$P(z|\xi) = \sum_{u=1}^U \eta_u f(z|\xi_u) \quad (1)$$

where z denotes the observation data, η_u denotes the positive weight coefficient that satisfies $\sum_{u=1}^U \eta_u = 1$, $\xi_u = (\mu_u, \sigma_u^2)$, and $f(z|\xi_u)$ denotes the Gaussian density function, which is formulated as

$$f(z|\xi_u) = \frac{1}{\sqrt{2\pi}\sigma_u} \exp\left(-\frac{(z - \mu_u)^2}{2\sigma_u^2}\right) \quad (2)$$

where μ_u is the mean of Gaussian distribution and σ_u is the standard deviation of Gaussian distribution. Then, the expectation-maximization (EM) method is applied for acquiring the distribution parameters of GMM. It assumes that the observation data are $z = (z_1, z_2, \dots, z_N)$. In the expectation step (E-step), the responsiveness of the u th component of

TABLE I
PARAMETER ESTIMATION PROCEDURE OF GMM

Algorithm 1: Parameter estimation process of GMM.

Input: The observed data, the constructed GMM.

Output: The parameters of GMM.

1. The parameters of GMM are first initialized randomly.
2. **E step:** According to the current model parameters, the responsiveness of components is calculated according to Eq. (3).
3. **M step:** Update the model parameters for the next iteration according to Eqs. (4), (5), and (6).
4. Execute steps 2 and 3 iteratively until the log-likelihood function of model converges.

GMM to the observed data z_i can be computed via the current parameters and it is defined as follows:

$$\hat{\gamma}_{iu} = \eta_u f(z_i|\xi_u) \bigg/ \sum_{u=1}^U \eta_u f(z_i|\xi_u) \quad i = 1, 2, \dots, N; \quad u = 1, 2, \dots, U. \quad (3)$$

In the maximization step (M-step), the parameters of GMM are updated using the responsiveness $\hat{\gamma}_{iu}$. The updating equations are defined as follows:

$$\hat{\mu}_u = \sum_{i=1}^N \hat{\gamma}_{iu} z_i \bigg/ \sum_{i=1}^N \hat{\gamma}_{iu} \quad (4)$$

$$\hat{\sigma}_u^2 = \sum_{i=1}^N \hat{\gamma}_{iu} (z_i - \hat{\mu}_u)^2 \bigg/ \sum_{i=1}^N \hat{\gamma}_{iu} \quad (5)$$

$$\hat{\eta}_u = \sum_{i=1}^N \hat{\gamma}_{iu} \bigg/ N. \quad (6)$$

The parameter estimation procedure of GMM is shown in Table I.

B. Exponential Mixture Model

The data distribution of time-domain signals can be estimated using GMM. Nonetheless, as mentioned earlier, the time-domain signals cannot fully reflect the degradation process of mechanical equipment. In certain cases, frequency-domain signals can better reflect the degradation trend. Therefore, it is necessary to extract degradation characteristics from the frequency domain. According to a number of distribution tests, the frequency data of mechanical vibration signals primarily contain exponential distribution characteristics. Thus, in this work, an EMM is designed to extract distribution characteristics from the frequency domain. The density function of EMM can be defined as

$$P(z|\zeta) = \sum_{r=1}^R \eta_r g(z|\zeta_r) \quad (7)$$

where $\zeta_r = \lambda_r$, $g(z|\zeta_r) = \lambda_r e^{-\lambda_r z}$, and $\sum_{r=1}^R \eta_r = 1$, $\eta_r > 0$. Based on the density function of EMM, the Gaussian components in GMM are replaced by the exponential components in EMM. Thereupon, EMM is capable of extracting distribution characteristics in the frequency domain from the

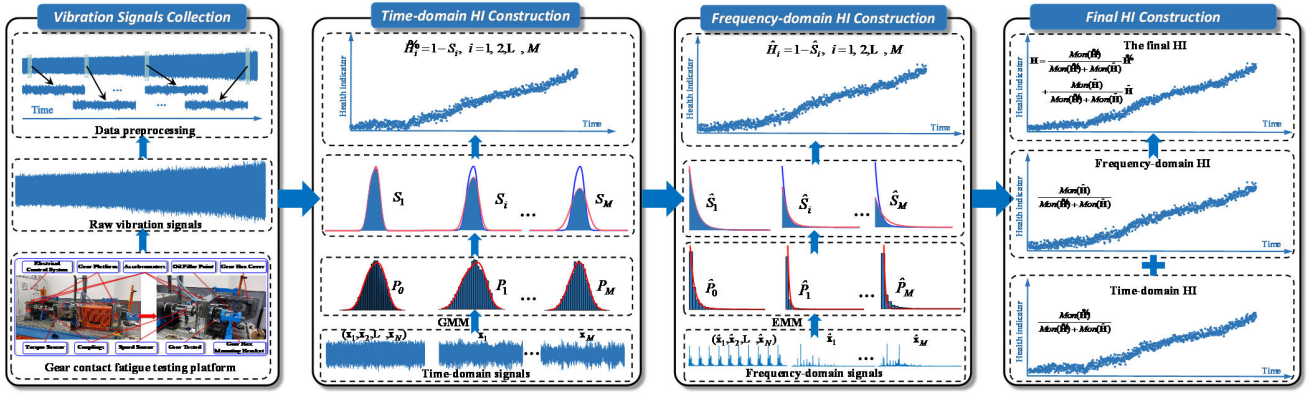


Fig. 1. Framework of the developed approach for constructing HI.

TABLE II
PARAMETER ESTIMATION PROCEDURE OF EMM

Algorithm 2: Parameter estimation process of EMM.

Input: The FFT data of vibration signals, the constructed EMM.

Output: The parameters of EMM.

1. The parameters of EMM are first initialized randomly.
2. **E step:** According to the current model parameters, the responsiveness of components is calculated according to Eq. (8).
3. **M step:** Update the model parameters for the next iteration according to Eqs. (9) and (10).
4. Execute steps 2 and 3 iteratively until the log-likelihood function of model converges.

exponential distribution perspective. The parameters of EMM are also obtained using the EM method. It assumes that the frequency-domain data of the sampled vibration signals are denoted by $z = (z_1, z_2, \dots, z_K)$. In the E-step, the responsiveness of the r th component of EMM to the observed data z_j can be calculated by the current parameters and it is defined as follows:

$$\hat{\gamma}_{jr} = \eta_r g(z_j | \zeta_r) / \sum_{r=1}^R \eta_r g(z_j | \zeta_r). \quad (8)$$

In the M-step, the model parameters of EMM are updated for the next iteration by the following equations:

$$\hat{\lambda}_r = \sum_{j=1}^K \hat{\gamma}_{jr} / \sum_{j=1}^K \hat{\gamma}_{jr} z_j \quad (9)$$

$$\hat{\eta}_r = \sum_{j=1}^K \hat{\gamma}_{jr} / K. \quad (10)$$

The parameter estimation procedure of EMM is presented in Table II.

C. Proposed-HI Construction Method

The Proposed-HI construction method comprises four main steps. In the first step, the vibration signals of the mechanical system are collected via multiple accelerometers. The raw vibration signals are preprocessed to reduce the noise in the acquired data. Moreover, frequency data are obtained by the FFT method. In the second step, the GMM is constructed

to estimate the data distributions of the processed vibration signals. DCRM is used to construct an HI based on the data distributions estimated from the time-domain perspective. In the third step, the EMM is built for frequency data distribution estimation. Subsequently, DCRM is also applied to generate an HI with the data distributions obtained from the frequency-domain perspective. In the final step, the final HI is generated by linearly weighting the HIs acquired in the above steps through monotonicity. A framework of the developed approach for constructing HI is exhibited in Fig. 1.

The developed approach for generating HI can be summarized by the following specific steps.

1) *Vibration Signal Collection:* Raw vibration signals of the mechanical system are collected via accelerometers under several operation conditions. Subsequently, processed vibration signals are obtained using a denoising algorithm, and the FFT method is utilized to acquire the FFT data of the vibration signals. The processed full life-cycle data are $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M)^T$ and $\mathbf{x}_i = (x_1, x_2, \dots, x_m)$, where M is the amount of the full life-cycle data and m is the data length of a single vibration signal. The full FFT data are $\hat{\mathbf{X}} = (\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \dots, \hat{\mathbf{x}}_M)^T$ and $\hat{\mathbf{x}}_i = (\hat{x}_1, \hat{x}_2, \dots, \hat{x}_n)$, where n is the data length of one FFT dataset.

2) *Time-Domain HI Construction:* GMM is constructed by setting the number of Gaussian components as K . The BIC is a criterion used for model selection, particularly in the fields of statistics and machine learning. It aims to balance the goodness of fit and the complexity of model to determine the most suitable model. Therefore, the number of Gaussian distributions and Student's t-distributions is set using the BIC, which is defined as follows:

$$\text{BIC} = k \ln(n) - 2 \ln(\hat{L}) \quad (11)$$

where k is the number of parameters to be estimated, n is the sample size, and $\hat{L}$ is the maximum value of the model's likelihood function. Next, the first N health data are utilized to estimate the baseline distribution P_0 , while the data distribution $\mathbf{P} = (P_1, P_2, \dots, P_M)$ for the entire life-cycle data $\mathbf{X}$ are separately estimated by the constructed GMM. Subsequently, the DCRM value $\tilde{H}_i$ in the time domain is

calculated based on P_0 and $\mathbf{P} = (P_1, P_2, \dots, P_M)$

$$\text{DCRM}(P_0 \| P_i) = \tilde{H}_i = 1 - S_i, \quad i = 1, 2, \dots, M \quad (12)$$

where S_i represents the area of the overlapping region between P_0 and P_i . It can be calculated by the following formula:

$$S_i = \int \min[P_0(x), P_i(x)] dx. \quad (13)$$

The DCRM is an indicator for measuring the distribution discrepancy, which satisfies all the requirements of distance measurement, including nonnegativity, symmetry, and triangle inequality. More theoretical explanations for the DCRM metric can be found in [29]. Finally, the HI vector $\tilde{\mathbf{H}} = (\tilde{H}_1, \tilde{H}_2, \dots, \tilde{H}_M)$ in the time domain is generated.

3) *Frequency-Domain HI Construction*: EMM is built by setting the number of exponential components as L . The number of exponential components is also determined by BIC. Next, the first N FFT data are utilized to estimate the baseline distribution $\hat{P}_0$, while the data distribution $\hat{\mathbf{P}} = (\hat{P}_1, \hat{P}_2, \dots, \hat{P}_M)$ for the entire FFT data $\hat{\mathbf{X}}$ is estimated using the EMM. The DCRM value $\hat{H}_i$ in the frequency domain is determined as follows:

$$\text{DCRM}(\hat{P}_0 \| \hat{P}_i) = \hat{H}_i = 1 - \hat{S}_i, \quad i = 1, 2, \dots, M \quad (14)$$

where $\hat{S}_i$ is the area of the overlapping region between $\hat{P}_0$ and $\hat{P}_i$. It can be calculated by the following formula:

$$\hat{S}_i = \int \min[\hat{P}_0(x), \hat{P}_i(x)] dx. \quad (15)$$

Finally, the HI vector $\hat{\mathbf{H}} = (\hat{H}_1, \hat{H}_2, \dots, \hat{H}_M)$ in the frequency domain is obtained.

4) *Final HI Construction*: Monotonicity (*Mon*) is a significant metric used to evaluate the performance of HIs. Thus, the *Mon* of the HIs in steps 2 and 3 is employed to generate the final HI, which can be calculated as follows:

$$\mathbf{H} = \frac{\text{Mon}(\tilde{\mathbf{H}})}{\text{Mon}(\tilde{\mathbf{H}}) + \text{Mon}(\hat{\mathbf{H}})} \tilde{\mathbf{H}} + \frac{\text{Mon}(\hat{\mathbf{H}})}{\text{Mon}(\tilde{\mathbf{H}}) + \text{Mon}(\hat{\mathbf{H}})} \hat{\mathbf{H}}. \quad (16)$$

The definition of *Mon* is provided in Section III. It should be noted that the final HI can comprehensively represent the health status and degradation trend of a machine from two domains, including time and frequency domains. Moreover, the generated HI is efficient for getting accurate RUL prediction results due to the favorable *Mon* of HI.

III. EXPERIMENTAL VERIFICATION AND COMPARISON

In this section, the performance of the developed approach is evaluated using several gear datasets. The constructed HIs of the gears are effectively constructed to quantitatively evaluate their degradation process based on the obtained datasets. Furthermore, comparative experiments confirm the effectiveness and superiority of the developed methodology.

A. Experimental Data Description

1) *Gear Dataset 1*: The vibration signals contained in this dataset were collected by a gear life testing platform at Newcastle University, U.K. The testing platform is exhibited in Fig. 2. It mainly comprises the control system, the test

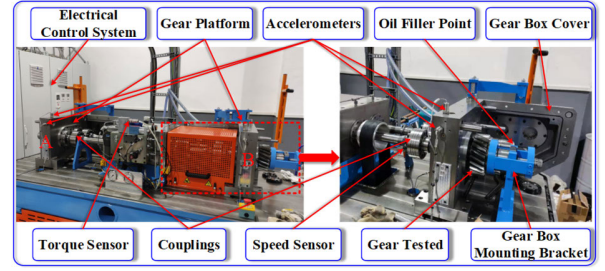


Fig. 2. Gear life testing platform for gear dataset 1.

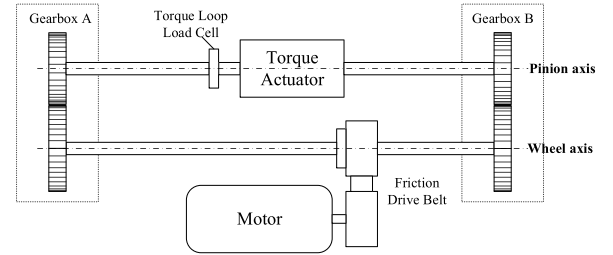


Fig. 3. Structural schematic of gear life testing platform.

TABLE III
DATA DESCRIPTION OF GEAR DATASET 1

Operating condition	Condition 1	Condition 2
Rotating speed (rpm)	1100	800
Torque (Nm)	4400	4070
Gearbox A	GearA1-1	GearA2-1
	GearA1-2	GearA2-2
Gearbox B	GearB1-1	GearB2-1
	GearB1-2	GearB2-2

platform, and the data acquisition system. In this platform, there are two gearboxes (i.e., gearboxes A and B), and the gears in them can be simultaneously tested under the same operating conditions. The structural schematic of gear life testing platform is shown in Fig. 3. Obviously, there are two gearboxes A and B. The center distance of gear pair in this testing platform is 160 mm. The gear material is 20Cr2Ni4A, and the module of gear is 6.154. The teeth numbers of two testing gears are 27 and 25. The vibration signals from gearboxes A and B were sampled at a frequency of 25 kHz, with each sample lasting for 0.2 s. The sampling interval was 1 min. The vibration signals of gears under two different operating conditions were collected, and the experimental description is illustrated in Table III.

2) *Gear Dataset 2*: Three run-to-failure datasets of gears were obtained by another gear life testing platform provided by Strama-MPS Inc., which is displayed in Fig. 4. There are four main components in this platform. The structural schematic of gearbox is shown in Fig. 5. The center distance of gear pair is set as 140 mm. The gear material is 40Cr, and the module of gear is 5. The teeth numbers of gears 1, 2, 3, and 4 are, respectively, 31, 25, 25, and 31. During the life tests, the life-cycle data were acquired through the installed sensors. The data description of collected gear datasets is listed in Table IV. The life-cycle data of all datasets were sampled at a frequency of 50 kHz. With regard to Subset 1, each sample

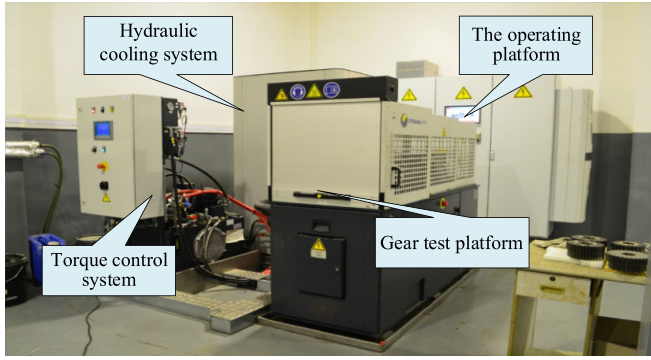


Fig. 4. Gear life testing platform for gear dataset 2.

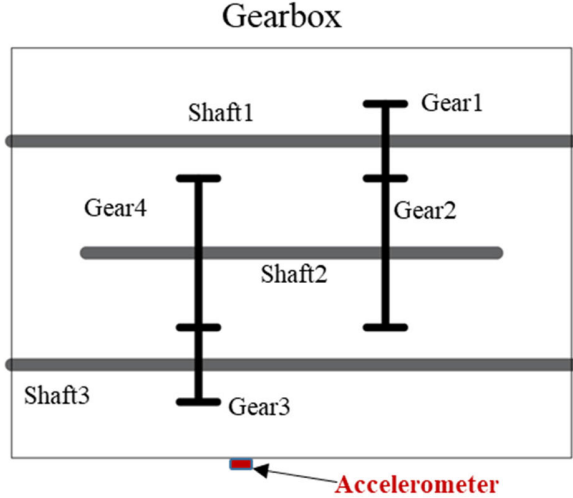


Fig. 5. Structural schematic of gearbox.

TABLE IV
DATA DESCRIPTION OF GEAR DATASET 2

Test parameter	Subset 1	Subset 2	Subset 3
Rotating speed (rpm)	500	1000	1000
Torque (Nm)	1400	1300	1300
The number of samples	600	600	600

lasts for 5 s with a collection interval of 30 s. For Subsets 2 and 3, the sampling duration and period were 10 s and 1 min, respectively. The vibration signals acquired during the steady stage exhibited a high degree of consistency. Consequently, they were not particularly useful for life prediction. To this end, only the final 600 samples were utilized in this work. These two gear datasets are available for downloading from the website.¹

B. Evaluation Metrics

Three classical evaluation metrics, i.e., *Mon*, correlation (*Corr*), and robustness (*Rob*), are utilized to quantitatively evaluate the performance of the obtained HIs of gears. First, the polynomial fitting method is used to decompose the HI into an average term and a random term

$$H(t_c) = H_T(t_c) + H_R(t_c) \quad (17)$$

where $H(t_c)$ is the HI value at time t_c , $H_T(t_c)$ is the average trend term, and $H_R(t_c)$ is the random term.

¹<https://qinyi-team.github.io/2024/01/Gear-life-cycle-data-download/>

The *Mon* is able to evaluate the quality of the upward and downward trend of the HI vector and is defined as follows:

$$Mon(\mathbf{H}) = \frac{1}{C-1} |\text{No. of } dH_T > 0 - \text{No. of } dH_T < 0| \quad (18)$$

where dH_T is the differential of H_T and C is the length of the HI vector.

Corr, which is calculated using the Pearson coefficient, can be utilized to evaluate the time correlation of the HI vector. It is defined as follows:

$$Corr(\mathbf{H}, \mathbf{T}) = \frac{\left| \sum_{c=1}^C (H(t_c) - \bar{H}) \times (t_c - \bar{T}) \right|}{\sqrt{\sum_{c=1}^C (H(t_c) - \bar{H})^2 \times \sum_{c=1}^C (t_c - \bar{T})^2}} \quad (19)$$

where $\bar{H} = (1/C) \sum_{c=1}^C H(t_c)$ and $\bar{T} = (1/C) \sum_{c=1}^C t_c$.

Rob is capable of estimating the tolerance of the HI vector to outliers and is defined as follows:

$$Rob(\mathbf{H}) = \exp\left(-\frac{\lambda}{C} \sum_{c=1}^C \left| \frac{H_R(t_c)}{\max(\mathbf{H}) - \min(\mathbf{H})} \right| \right) \quad (20)$$

where λ is a relaxation factor that can control the strictness of *Rob*.

Finally, a composite index (*CI*) that combines the above evaluation metrics is defined as follows:

$$CI = v_1 \times Mon + v_2 \times Corr + v_3 \times Rob \quad (21)$$

where v is a positive value that satisfies $\sum_{i=1}^3 v_i = 1$. In this work, $v_1 = 0.4$, $v_2 = 0.3$, and $v_3 = 0.3$. Unlike some articles that may assign equal weights, we think that *Mon*, *Corr*, and *Rob* play different roles in evaluating the performance of HI on RUL prediction. It follows that they are not considered equally important in this work. The *Mon* of HI is more significant for the RUL prediction compared to *Corr* and *Rob*, and hence, *Mon* is assigned a maximum weight, while *Corr* and *Rob* are assigned relatively small weights [34].

C. HI Construction Based on the Developed Approach

To validate the effectiveness and feasibility of the developed approach for constructing HI, the vibration signals of GearA1-1, GearA1-2, and GearB1-1 in gear dataset 1 are employed. In the time-domain HI construction step, the number of Gaussian distributions in GMM is determined by BIC, and 20 is selected as the maximum number of iterations for the EM algorithm. In addition, the initialization parameters of GMM are obtained using the *kmeans2* approach. In the frequency-domain HI construction step, the number of exponential distributions in EMM is also determined by BIC and the maximum number of parameter updating is also selected to 20. The data distributions of the vibration signals are estimated by the constructed GMM, and those of the FFT data are acquired by the constructed EMM. Subsequently, the time and frequency domain HIs are generated by DCRM. Finally, the final HIs of the gears are obtained by linearly weighting the above HIs through monotonicity. Fig. 4 displays the constructed HIs from the developed approach, indicating

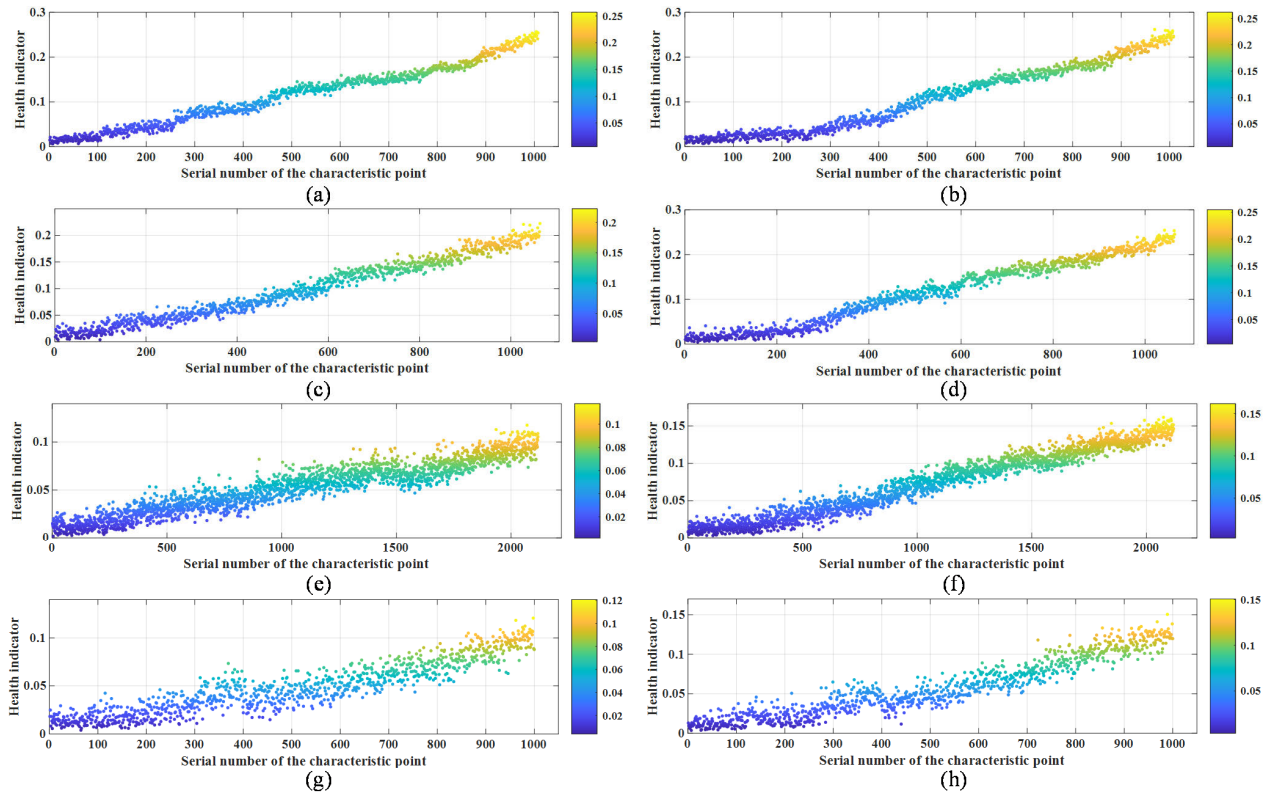


Fig. 6. Constructed gear HIs of gear dataset 1. (a) GearA1-1. (b) GearA1-2. (c) GearB1-1. (d) GearB1-2. (e) GearA2-1. (f) GearA2-2. (g) GearB2-1. (h) GearB2-2.

that the HIs constructed based on the proposed method have a monotonous upward trend and can effectively represent the gear's deterioration trends over time. In the preliminary stage, the HI value is close to zero, indicating that the distribution discrepancy between current and health data is small. This suggests that the tested gear is currently in a healthy state, and no faults are observed on its tooth surface. As the running time elapses, the HI value becomes gradually larger, indicating that the distribution discrepancy becomes larger. Consequently, different types of gear faults will occur on the gear surface. Finally, the HIs reach their maximum values, illustrating that the gear cannot function properly due to certain severe faults.

All subsets in gear dataset 2 are also employed to further validate the efficiency and feasibility of the developed approach. Fig. 5 displays the results of HI construction. Similarly, the obtained HIs appear appropriate to quantitatively describe the degradation trend with running time. Overall, according to Figs. 6 and 7, the developed method for constructing HI is effective and feasible to construct gear HIs and identify their deterioration process.

D. Comparison and Ablation Experiments

In contrast experiment, the benchmark methods, including rms, Kurtosis [35], gini index (GI) [36], and approximate entropy (ApEn) [37], and the classical and advanced unsupervised HI construction methods, including PCA, KPCA, ISOMAP [21], GMM with Kullback–Leibler divergence (GKLD) [38], and GMM with confidence value (GCV) [39], are employed for comparison to the Proposed-HI construction approach. Moreover, HI construction approaches that construct

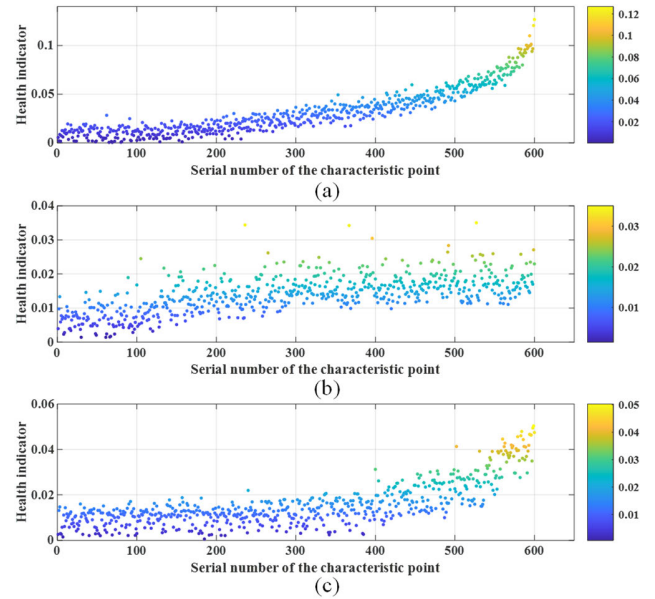


Fig. 7. Constructed gear HIs of gear dataset 2. (a) Subset 1. (b) Subset 2. (c) Subset 3.

the HI via GMM [29] and EMM are also employed for ablation experiment. With respect to PCA, KPCA, and ISOMAP, ten features, which are calculated from time and frequency domains, are used as the input data. Subsequently, gear HIs are generated by fusing these extracted features through PCA, KPCA, and ISOMAP. All the evaluation metrics described in this section are used to quantitatively evaluate the performance of different HIs. The results of these evaluation metrics for

TABLE V
EVALUATION RESULTS OF DIFFERENT GEAR HIS

Method	GearA1-1					GearA1-2					GearB1-1				
	Mon	Corr	Rob	CI	PR	Mon	Corr	Rob	CI	PR	Mon	Corr	Rob	CI	PR
RMS	0.678	0.963	0.956	0.847	↑3.2%	0.605	0.978	0.949	0.820	↑3.5%	0.219	0.902	0.848	0.613	↑33.8%
Kurtosis	0.017	0.606	0.786	0.424	↑106.1%	0.150	0.779	0.779	0.527	↑61.1%	0.057	0.512	0.617	0.361	↑127.1%
GI	0.047	0.429	0.767	0.378	↑131.2%	0.136	0.705	0.760	0.494	↑71.9%	0.001	0.068	0.433	0.151	↑443.0%
ApEn	0.223	0.663	0.807	0.530	↑64.9%	0.061	0.019	0.860	0.288	↑194.8%	0.334	0.958	0.870	0.682	↑20.2%
PCA	0.071	0.717	0.812	0.487	↑79.5%	0.200	0.843	0.800	0.573	↑48.2%	0.212	0.896	0.845	0.607	↑35.1%
KPCA	0.078	0.723	0.815	0.493	↑77.3%	0.194	0.846	0.801	0.572	↑48.4%	0.208	0.900	0.845	0.607	↑35.1%
ISOMAP	0.221	0.884	0.835	0.604	↑44.7%	0.313	0.925	0.860	0.661	↑28.4%	0.225	0.924	0.857	0.624	↑31.4%
GKLD	0.706	0.929	0.958	0.848	↑3.1%	0.636	0.941	0.950	0.822	↑3.3%	0.274	0.781	0.864	0.603	↑36.0%
GCV	0.543	0.918	0.942	0.775	↑12.8%	0.557	0.938	0.938	0.786	↑8.0%	0.281	0.769	0.869	0.604	↑35.8%
GMM	0.646	0.976	0.950	0.836	↑4.6%	0.636	0.976	0.936	0.828	↑2.5%	0.289	0.824	0.837	0.614	↑33.5%
EMM	0.605	0.983	0.933	0.817	↑7.0%	0.635	0.981	0.940	0.830	↑2.3%	0.590	0.986	0.928	0.810	↑1.2%
Proposed	0.726	0.989	0.956	0.874	/	0.672	0.982	0.951	0.849	/	0.612	0.985	0.931	0.820	/
Method	GearB1-2					GearA2-1					GearA2-2				
	Mon	Corr	Rob	CI	PR	Mon	Corr	Rob	CI	PR	Mon	Corr	Rob	CI	PR
RMS	0.445	0.972	0.930	0.749	↑13.1%	0.218	0.947	0.865	0.631	↓3.9%	0.336	0.981	0.922	0.705	↓2.7%
Kurtosis	0.007	0.684	0.755	0.434	↑95.2%	0.130	0.830	0.778	0.534	↑13.5%	0.041	0.211	0.432	0.209	↑228.2%
GI	0.076	0.781	0.763	0.494	↑71.5%	0.077	0.674	0.651	0.428	↑41.6%	0.017	0.092	0.518	0.190	↑261.1%
ApEn	0.479	0.968	0.904	0.753	↑12.5%	0.002	0.605	0.722	0.399	↑51.9%	0.004	0.012	0.584	0.180	↑281.1%
PCA	0.240	0.907	0.851	0.623	↑36.0%	0.161	0.904	0.828	0.584	↑3.8%	0.076	0.624	0.633	0.407	↑68.6%
KPCA	0.244	0.910	0.854	0.627	↑35.1%	0.169	0.895	0.819	0.582	↑4.1%	0.067	0.650	0.641	0.414	↑65.7%
ISOMAP	0.283	0.925	0.864	0.650	↑30.3%	0.178	0.911	0.828	0.593	↑2.2%	0.097	0.854	0.766	0.525	↑30.7%
GKLD	0.430	0.936	0.926	0.731	↑15.9%	0.170	0.923	0.856	0.602	↑0.7%	0.300	0.948	0.926	0.682	↑0.6%
GCV	0.396	0.940	0.903	0.711	↑19.1%	0.097	0.835	0.798	0.529	↑14.6%	0.243	0.942	0.902	0.650	↑5.5%
GMM	0.415	0.966	0.911	0.729	↑16.2%	0.103	0.919	0.828	0.565	↑7.3%	0.289	0.978	0.912	0.683	↑0.4%
EMM	0.680	0.988	0.940	0.850	↓0.3%	0.136	0.920	0.854	0.587	↑3.2%	0.177	0.920	0.833	0.597	↑14.9%
Proposed	0.671	0.987	0.943	0.847	/	0.154	0.944	0.871	0.606	/	0.297	0.978	0.912	0.686	/
Method	GearB2-1					GearB2-2					Subset 1				
	Mon	Corr	Rob	CI	PR	Mon	Corr	Rob	CI	PR	Mon	Corr	Rob	CI	PR
RMS	0.375	0.918	0.844	0.679	↓4.4%	0.495	0.960	0.895	0.754	↓6.4%	0.075	0.087	0.585	0.232	↑240.9%
Kurtosis	0.149	0.250	0.551	0.300	↑116.3%	0.043	0.460	0.703	0.366	↑92.9%	0.469	0.794	0.882	0.690	↑14.6%
GI	0.051	0.238	0.499	0.241	↑169.3%	0.063	0.390	0.653	0.338	↑108.9%	0.078	0.235	0.662	0.300	↑163.7%
ApEn	0.021	0.055	0.294	0.113	↑474.3%	0.189	0.548	0.730	0.459	↑53.8%	0.412	0.917	0.805	0.681	↑16.1%
PCA	0.207	0.635	0.707	0.485	↑33.8%	0.115	0.717	0.716	0.476	↑48.3%	0.623	0.868	0.914	0.784	↑0.9%
KPCA	0.215	0.639	0.702	0.488	↑33.0%	0.119	0.725	0.718	0.480	↑47.1%	0.629	0.881	0.913	0.790	↑0.1%
ISOMAP	0.219	0.662	0.714	0.500	↑29.8%	0.183	0.846	0.789	0.564	↑25.2%	0.656	0.875	0.926	0.803	↓1.5%
GKLD	0.317	0.892	0.848	0.649	↑0.0%	0.471	0.910	0.901	0.732	↓3.5%	0.289	0.324	0.800	0.453	↑74.6%
GCV	0.279	0.840	0.837	0.615	↑5.5%	0.339	0.872	0.898	0.667	↑5.9%	0.192	0.354	0.728	0.401	↑97.3%
GMM	0.279	0.901	0.826	0.630	↑3.0%	0.433	0.944	0.887	0.722	↓2.2%	0.242	0.388	0.648	0.408	↑93.9%
EMM	0.189	0.820	0.757	0.549	↑18.2%	0.155	0.734	0.697	0.491	↑43.8%	0.613	0.896	0.904	0.785	↑0.8%
Proposed	0.299	0.918	0.846	0.649	/	0.399	0.944	0.877	0.706	/	0.629	0.893	0.904	0.791	/
Method	Subset 2					Subset 3					Average results				
	Mon	Corr	Rob	CI	PR	Mon	Corr	Rob	CI	PR	Mon	Corr	Rob	CI	PR
RMS	0.022	0.361	0.513	0.271	↑92.6%	0.239	0.780	0.833	0.580	↑6.7%	0.337	0.804	0.831	0.625	↑15.8%
Kurtosis	0.122	0.117	0.404	0.205	↑154.6%	0.315	0.745	0.897	0.619	↑0.0%	0.136	0.544	0.689	0.424	↑70.8%
GI	0.067	0.183	0.525	0.239	↑118.4%	0.179	0.592	0.837	0.500	↑23.8%	0.072	0.399	0.642	0.341	↑112.3%
ApEn	0.421	0.859	0.769	0.657	↓20.5%	0.486	0.909	0.855	0.724	↓14.5%	0.239	0.592	0.745	0.497	↑45.7%
PCA	0.105	0.246	0.456	0.253	↑106.3%	0.275	0.762	0.869	0.599	↑3.3%	0.208	0.738	0.766	0.534	↑35.6%
KPCA	0.108	0.248	0.452	0.253	↑106.3%	0.275	0.768	0.868	0.601	↑3.0%	0.210	0.744	0.766	0.537	↑34.8%

different methods are listed in Table V. To further highlight the superiority of the developed HI construction method, the promoting rate (PR) of the proposed method compared to other methods is calculated and given in Table V. The PR is defined

as

$$PR = \left(\frac{CI_{\text{Proposed method}} - CI_{\text{Comparative method}}}{CI_{\text{Comparative method}}} \right) \times 100\%. \quad (22)$$

TABLE V
(Continued.) EVALUATION RESULTS OF DIFFERENT GEAR HIS

ISOMAP	0.092	0.257	0.449	0.249	↑109.6%	0.359	0.762	0.877	0.635	↓2.5%	0.257	0.802	0.797	0.582	↑24.4%
GKLD	0.262	0.450	0.553	0.406	↑28.6%	0.242	0.742	0.896	0.588	↑5.3%	0.372	0.798	0.862	0.647	↑11.9%
GCV	0.269	0.541	0.574	0.442	↑18.1%	0.165	0.724	0.817	0.528	↑17.2%	0.305	0.788	0.837	0.610	↑18.7%
GMM	0.275	0.577	0.624	0.470	↑11.1%	0.172	0.768	0.806	0.541	↑14.4%	0.343	0.838	0.833	0.639	↑13.3%
EMM	0.195	0.560	0.557	0.413	↑26.4%	0.372	0.691	0.794	0.594	↑4.2%	0.395	0.862	0.831	0.666	↑8.7%
Proposed	0.372	0.638	0.607	0.522	/	0.386	0.733	0.816	0.619	/	0.474	0.908	0.874	0.724	/

According to Table V, the developed approach has better evaluation results compared to other approaches. Taking GearA1-1 as an example, the PR results demonstrate that the *CI* of Proposed-HI is 3.2%, 106.1%, 131.2%, 64.9%, 79.5%, 77.3%, 44.7%, 3.1%, 12.8%, 4.6%, and 7.0% higher than those of rms, Kurtosis, GI, ApEn, PCA-based HI, KPCA-based HI, ISOMAP-based HI, GKLD-based HI, GCV-based HI, GMM-based HI, and EMM-based HI, respectively. For GearA2-1, GearA2-2, and GearB2-1, the proposed method is inferior to rms, but it outperforms other methods. For GearB2-2, the *CI* of the presented approach is lower than those of rms, GKLD, and GMM, but it is higher than those of Kurtosis, GI, ApEn, PCA, KPCA, ISOMAP, GCV, and EMM. For Subset 3, the *CI* of Proposed-HI is lower than those of ApEn and ISOMAP, but it is higher than those of rms, GI, PCA, KPCA, GKLD, GCV, GMM, and EMM. It can also be known from Table V that the monotonic tendency of Proposed-HI is better than other methods. The favorable monotonic performance of HI is beneficial to RUL prediction. Taking GearA1-1 as an example, the *Mon* of Proposed-HI is 0.726, which is higher than those of rms, Kurtosis, GI, ApEn, PCA-based HI, KPCA-based HI, ISOMAP-based HI, GKLD-based HI, and GCV-based HI. Similar results can be concluded from the *Mon* values of GearA1-2, GearB1-1, and GearB1-2. This demonstrates that the constructed HI by the presented approach has the capacity to assess the gear deterioration process and has higher performance than other HI construction methods. Moreover, the *Mon* of Proposed-HI is higher than those of GMM-based HI and EMM-based HI for all gear datasets except for GearB1-2 and GearB2-2, suggesting that the Proposed-HI construction method that adaptively fuses time and frequency domain HIs is more appropriate for constructing the gear HI than both GMM and EMM. This is primarily because the Proposed-HI construction approach is able to extract degradation information from both time and frequency domains. In regard to *Corr* and *Rob*, the presented approach is superior to or comparable to the contrast approaches, indicating that Proposed-HI has a better temporal correlation and a higher outlier tolerance performance.

In order to further validate the superiority of Proposed-HI, the average results of *Mon*, *Corr*, *Rob*, and *CI* for the two gear datasets are also given in Table V. It can be seen from the average results that Proposed-HI has the highest *CI* among these 12 HIs, and it is 15.8%, 70.8%, 112.3%, 45.7%, 35.6%, 34.8%, 24.4%, 11.9%, 18.7%, 13.3%, and 8.7% higher than those of rms, Kurtosis, GI, ApEn, PCA-based HI, KPCA-based HI, ISOMAP-based HI, GKLD-based HI, GCV-based HI, GMM-based HI, and EMM-based HI, respectively. In addition, the *Mon*, *Corr*, and *Rob* values of Proposed-HI are

larger than those of contrast methods. It then can be concluded that Proposed-HI has higher monitoring performance, and it can be better applied to time series-based RUL prediction.

IV. APPLICATION TO GEAR RUL PREDICTION

The multihierarchical long-term memory augmented network (MLMA-Net) is first constructed for time series-based life prediction. Subsequently, the gear RUL is estimated via Proposed-HIs and the constructed MLMA-Net to verify that the developed HI is appropriate for life prediction. Finally, the superiority of Proposed-HI in life prediction is validated by comparison to other HIs.

A. MLMA-Net-Based RUL Prediction Method

After constructing HI vectors based on the gear monitoring datasets, an appropriate RUL prediction method should be applied to predict the gear RULs. Essentially, the time series-based RUL estimation approach is able to estimate unknown HI points by learning degradation information from known HI points. LSTM networks [40] have been broadly explored for RUL prediction of machinery. Moreover, several variants of LSTM have been designed to enhance their memory capacity. In particular, the MLMA-Net is able to realize RUL prediction using sequence information and some gating values close to the boundary regions. Furthermore, the availability of MLMA-Net for RUL estimation has been proved in the previous work. Therefore, we choose MLMA-Net as the prediction network to demonstrate that the presented HI is more appropriate for RUL prediction than other HIs. Detailed information on MLMA-Net can be found in [41] and its cell structure is presented in Fig. 8.

Based on the cell state $\mathbf{h}_t$ of MLMA-Net, its output can be calculated as follows:

$$\begin{aligned}
 \mathbf{y}_t^1 &= \text{Dropout}(\mathbf{h}_t) \\
 \mathbf{y}_t^2 &= \text{Dropout}(\text{Relu}(\mathbf{W}_1 \mathbf{y}_t^1 + \mathbf{b}_1)) \\
 \mathbf{y}_t^3 &= \mathbf{W}_2 \mathbf{y}_t^2 + \mathbf{b}_2 \\
 \mathbf{y}_t &= \mathbf{W}_3 (\text{Dropout}(\mathbf{y}_t^3 + \mathbf{h}_t)) + \mathbf{b}_3
 \end{aligned} \tag{23}$$

where $\mathbf{W}$ and $\mathbf{b}$ are updatable parameters and $\mathbf{y}_t$ is the output of the MLMA-Net.

The constructed HIs of GearA1-2 and Subset 2 are utilized for the gear RUL prediction. The learnable parameters of MLMA-Net are initialized by a uniform initialization method and updated by the stochastic gradient descent (SGD) algorithm. A flowchart of the MLMA-Net-based life estimation approach is exhibited in Fig. 9.

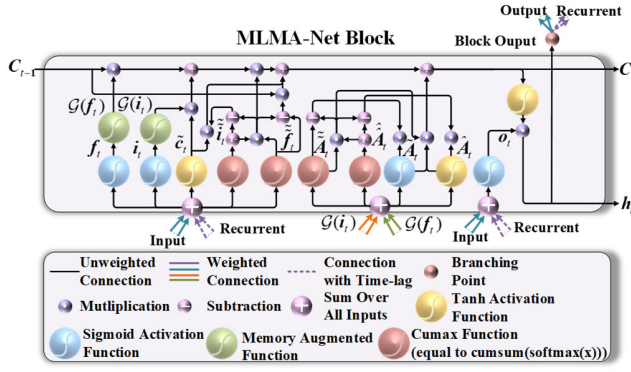


Fig. 8. Cell structure of MLMA-Net.

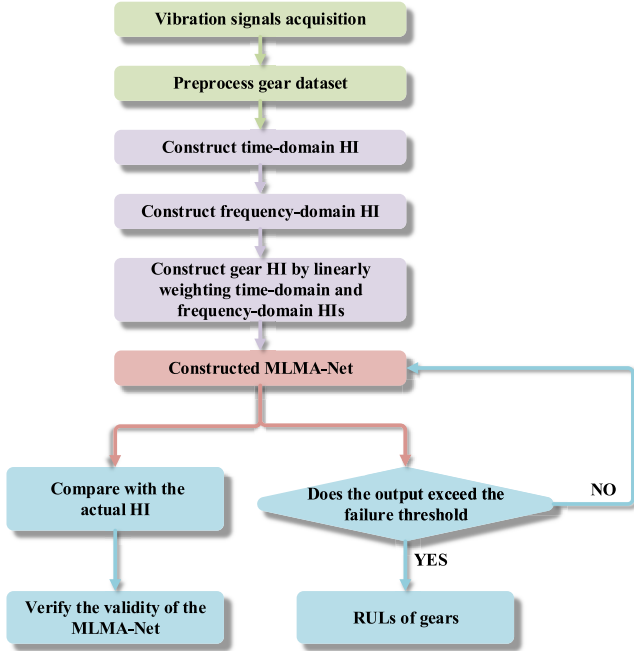


Fig. 9. Flowchart of the MLMA-Net-based life estimation approach.

B. RUL Prediction Using the Constructed HI

The HIs generated using the developed approach are utilized for predicting the gear RULs through the constructed MLMA-Net. As regards GearA1-2, the last 60 HI values are estimated using the training dataset, which is constructed by the first 948 HI values. As regards Subset 2, the MLMA-Net is trained via the first 540 HI values. Then, the remaining 60 HI values are estimated in the testing phase. Therefore, the actual RULs of GearA1-2 and Subset 2 are both equal to 60 min. The hyperparameters of MLMA-Net are shown in Table VI.

The RULs of GearA1-2 and Subset 2 are acquired via the trained MLMA-Net by employing the configured hyperparameters. To ensure the fairness of the predicted RULs, the trained MLMA-Net is tested 100 times on gear HI. The average of the 100 obtained RULs was then considered to be the final RUL. Furthermore, to quantify the uncertainty of MLMA-Net, the dropout operation was enabled during testing. The probability density functions (PDFs), 95% confidence interval, True RULs, and estimated RULs of GearA1-2 and Subset 2 are given in Table VII and Fig. 10. By analyzing the distributions of the RUL prediction results in some

TABLE VI
HYPERPARAMETERS OF MLMA-NET

Hyper-parameters	GearA1-2	Subset 2
Number of neurons		
Input layer	100	60
MLMA-Net cell	20	17
Output layer	1	1
Dropout rate	0.03	0.01
Optimizer	SGD	SGD
Training epoch	300	300

TABLE VII
RESULTS OF RUL PREDICTION FOR GEARS

	True RUL (min)	Estimated RUL (min)	95% confidence interval
GearA1-2	60	57.53	[47.77 67.29]
Subset 2	60	60.66	[20.23 101.09]

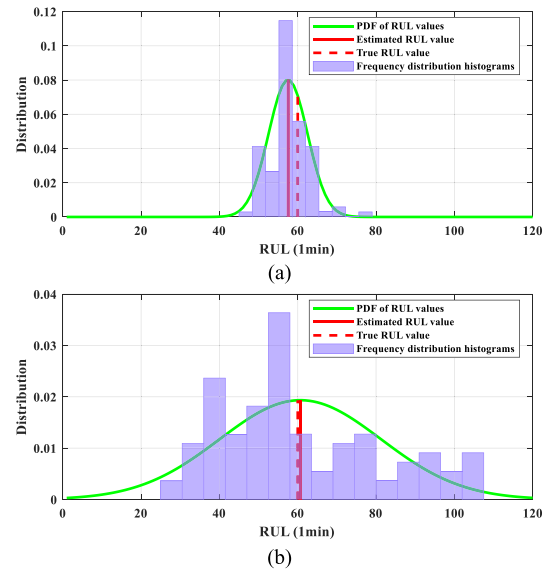


Fig. 10. Prediction results with uncertainty assessment. (a) GearA1-2. (b) Subset 2.

relevant literature [42], [43], [44], it can be found that the obtained RULs have the characteristics of normal distribution. Therefore, the normal distribution is considered here to fit the RUL prediction results. In this work, the distribution of RUL prediction results is fit by the MATLAB function “fitdist.” In addition, the standard deviation of the RUL distribution is also calculated by this function.

Table VII and Fig. 8 illustrate that the estimated RULs are close to the true ones, and their values are all within a 95% confidence interval. In addition, the estimated RULs of GearA1-2 and Subset 2 are all in the center of the probability distributions. Based on the obtained PDFs, proper control strategies and maintenance plans can be made to guarantee the efficient operation of mechanical equipment. The above results indicate that the constructed HI based on the developed approach is appropriate to predict gear RUL.

C. Comparison With Other HIs Used for RUL Prediction

The constructed HIs of GearA1-2 and Subset 2 based on GMM, EMM, GCV, GKLD, ISOMAP, KPCA, PCA, ApEn, GI, Kurtosis, and rms are all used for comparison to reveal

TABLE VIII

EVALUATION RESULTS OF DIFFERENT GEAR HIs FOR GEARA1-2

Method	MAE	MAPE	RMSE	NRMSE	Score
RMS	13.060	21.8%	14.495	0.305	0.480
Kurtosis [34]	21.440	35.7%	22.491	0.583	0.314
GI-HI [35]	23.820	39.7%	28.111	0.606	0.263
ApEn-HI [36]	26.360	43.9%	30.831	0.664	0.201
PCA-HI	23.100	38.5%	23.187	0.279	0.005
KPCA-HI	21.920	36.5%	22.571	0.593	0.295
ISOMAP-HI [21]	22.040	36.7%	22.804	0.601	0.299
GKLD-HI [37]	12.010	20.0%	12.073	0.252	0.501
GCV-HI [38]	9.660	16.1%	10.061	0.200	0.580
GMM-HI [29]	14.040	23.4%	15.134	0.327	0.456
EMM-HI	9.630	16.0%	11.823	0.172	0.300
Proposed-HI	4.350	7.3%	5.536	0.096	0.723

TABLE IX

EVALUATION RESULTS OF DIFFERENT GEAR HIs FOR SUBSET 2

Method	MAE	MAPE	RMSE	NRMSE	Score
RMS	28.130	46.9%	31.987	0.596	0.188
Kurtosis [34]	31.070	51.8%	36.227	0.709	0.223
GI-HI [35]	34.920	58.2%	37.366	1.342	0.156
ApEn-HI [36]	22.770	37.9%	27.770	0.599	0.254
PCA-HI	37.620	62.7%	44.256	0.975	0.202
KPCA-HI	34.080	56.8%	38.854	0.709	0.140
ISOMAP-HI [21]	28.780	48.0%	32.864	1.035	0.252
GKLD-HI [37]	21.310	35.5%	21.501	0.351	0.162
GCV-HI [38]	23.020	38.4%	26.484	0.662	0.257
GMM-HI [29]	27.640	46.1%	31.491	0.700	0.219
EMM-HI	24.800	41.5%	28.349	0.656	0.235
Proposed-HI	16.800	28.0%	20.535	0.338	0.357

the advantage of the presented approach. Five evaluation metrics, i.e., mean absolute error (MAE), mean absolute percentage error (MAPE), rms error (RMSE), normalized RMSE (NRMSE), and Score, are employed for quantitatively assessing the prediction performance of different gear HIs. Their comparative results for GearA1-2 and Subset 2 are demonstrated in Tables VIII and IX, respectively.

It can be known from Tables VIII and IX that Proposed-HI has the best evaluation results compared with other methods. In particular, the Proposed-HI is significantly superior to the ISOMAP-based HI (ISOMAP-HI), KPCA-based HI (KPCA-HI), and PCA-based HI (PCA-HI) in terms of MAE, MAPE, RMSE, NRMSE, and Score. This demonstrates that the prediction performance of the presented approach is better than that of feature fusion approaches, and the Proposed-HI is more efficient to estimate the gear RULs. Moreover, the Proposed-HI is better than the GMM-based HI (GMM-HI), EMM-based HI (EMM-HI), GKLD-based HI (GKLD-HI), GCV-based HI (GCV-HI), ApEn-based HI (ApEn-HI), GI-based HI (GI-HI), Kurtosis, and rms in terms of the five evaluation metrics, demonstrating that the presented method can generate better gear HIs for RUL prediction. Finally, it can be concluded that the Proposed-HI has the best prediction performance than other HIs.

V. CONCLUSION

Most of the existing HI construction methods are capable of constructing HIs from the time domain. However, time-domain signals cannot fully reflect the degradation trends of mechanical equipment, and in some cases, they contain fewer degradation characteristics than frequency-domain signals. To improve the applicability and quality of HI

under various conditions, in this work, a novel unsupervised HI construction approach based on GMM and EMM has been proposed from two domains. In this method, an EMM was first successfully employed to extract the frequency-domain distribution characteristics from frequency-domain data. Furthermore, a GMM was applied to mine time-domain distribution characteristics from time-domain data. Subsequently, time and frequency domain HIs were constructed by DCRM, which could quantify the discrepancies between the baseline distribution and the data distributions at different degradation moments. Considering the monotonicity of HIs from the two domains, the final gear HI was constructed by linearly weighting the obtained time and frequency domain HIs. The effectiveness of the developed HI construction method was validated through multiple gear life-cycle datasets. The RULs were accurately estimated via Proposed-HIs and MLMA-Net, and the uncertainty of the latter was quantified using the PDFs of RULs. Using the predicted RULs, certain proper control strategies and maintenance plans can be made for the safe and efficient operation of mechanical equipment. Finally, comparative results of gear RUL prediction experiments demonstrated that the presented approach is superior to other unsupervised HI construction approaches, and it is more appropriate to estimate the RULs of gears.

In the future work, it is worth exploring an unsupervised HI construction method based on mixture model under time-varying operation conditions. In addition, HIs for both early fault warning and subsequent RUL prediction need to be researched.

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